

Tear fluid osmolarity as a potential marker of hydration status.

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It has been suggested that tear fluid is isotonic with plasma, and plasma osmolality (P(osm)) is an accepted, albeit invasive, hydration marker. Our aim was to determine whether tear fluid osmolarity (T(osm)) assessed using a new, portable, noninvasive, rapid collection and measurement device tracks hydration. PURPOSE: This study aimed to compare changes in T(osm) and another widely used noninvasive marker, urine specific gravity (USG), with changes in P(osm) during hypertonic-hypovolemia. METHODS: In a randomized order, 14 healthy volunteers exercised in the heat on one occasion with fluid restriction (FR) until 1%, 2%, and 3% body mass loss (BML) and with overnight fluid restriction until 08:00 h the following day, and on another occasion with fluid intake (FI). Volunteers were rehydrated between 08:00 and 11:00 h. T(osm) was assessed using the TearLab osmolarity system. RESULTS: P(osm) and USG increased with progressive dehydration on FR ($P < 0.001$). T(osm) increased significantly on FR from 293 ± 9 to 305 ± 13 mOsm·L⁻¹ at 3% BML and remained elevated overnight (304 ± 14 mOsm·L⁻¹; $P < 0.001$). P(osm) and T(osm) decreased during exercise on FI and returned to preexercise values the following morning. Rehydration restored P(osm), USG, and T(osm) to within preexercise values. The mean correlation between T(osm) and P(osm) was $r = 0.93$ and that between USG and P(osm) was $r = 0.72$. CONCLUSIONS: T(osm) increased with dehydration and tracked alterations in P(osm) with comparable utility to USG. Measuring T(osm) using the TearLab osmolarity system may offer sports medicine practitioners, clinicians, and research investigators a practical and rapid hydration assessment technique.